

LAB MANUAL



**Faculty of Computing, Engineering
and The Built Environment**

COM682 – Lab 05

Azure REST APIs

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Prerequisites:

- [Install Git](#)
- [Install the latest .NET Core 3.1 SDK](#)
- [Install Python](#)
- [Install Postman](#)

In this lab, we will learn about hosting a web API on Azure and how to use it. This lab is based on two parts, and in the first part, we will go through API and how to use it. If you know about API, skip the first part, and in the second part, we will go through how to deploy web API on Azure and use it.

Part 1

1. What is API?

API stands for Application Programmable Interface. It is a connection between computer programs. Let us assume a web application that asks some questions from the user and returns the latest movie that the program thinks the user will like. The web application is written in JSON, and the server with all the data and models is written in Python. Web applications need some medium to communicate with the server and, most importantly, in a secure environment. To solve this issue, API is used. The web application will call an API, send the user answerd question, and send it to the server. In return, the server will return its predicated latest movie the user might like.

2. Using an API

To make yourself familiar, we will use StackOverflow API in PostMan. First, install the Postman by clicking the following link.

<https://www.postman.com/downloads/>

Download the Postman and install it. (You might still need to upgrade even if its installed)

After downloading and installing the Postman, go to the following StackOverflow API page.

<https://api.stackexchange.com/docs>

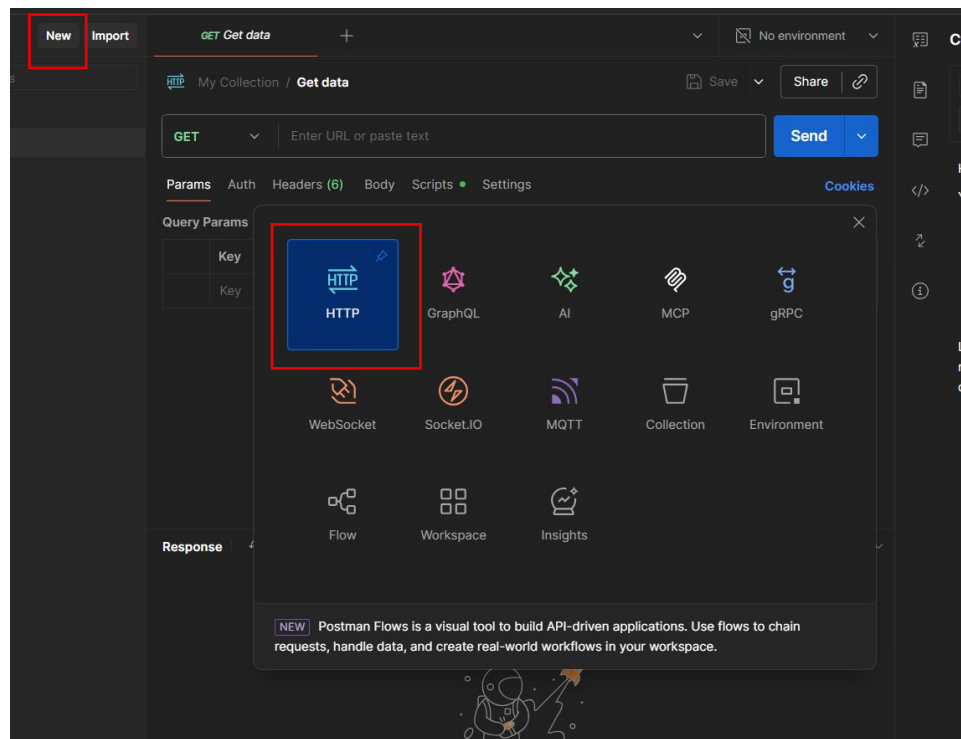
There is several APIs that can be used, but the most interesting ones are the question.

<https://api.stackexchange.com/docs/questions>

Open the Postman

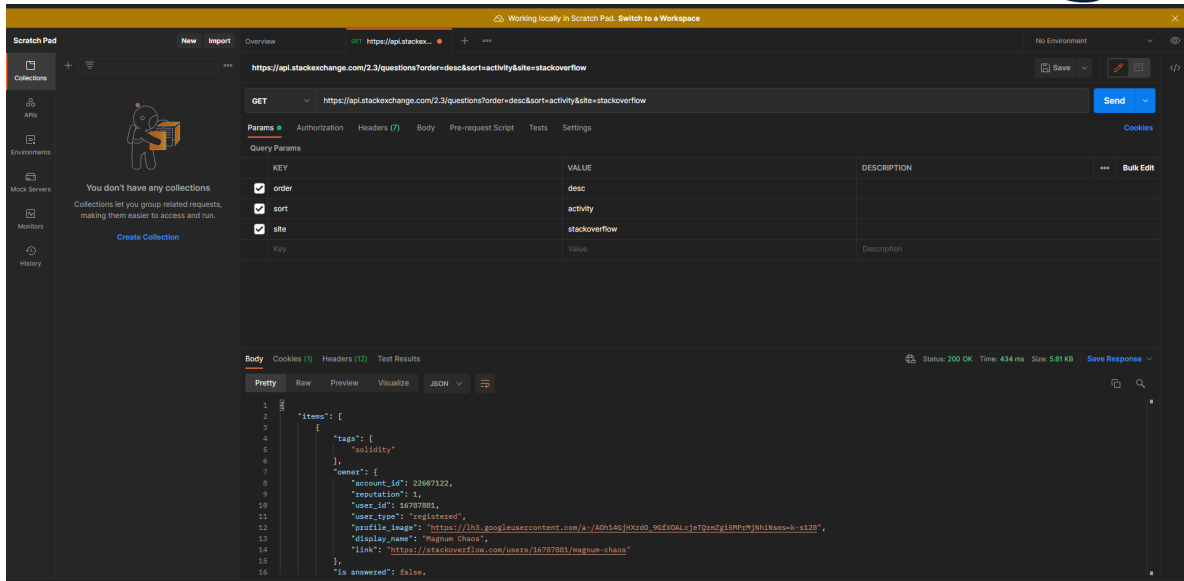
click on the file and click on the new file.

Click on the HTTP request.



Put the following link in the GET box and click on send.

<https://api.stackexchange.com/2.3/questions?order=desc&sort=activity&site=stackoverflow>



You can see that you have successfully engaged with API, and you can see all the questions on StackOverflow. You should see the code **200 OK**

3. API Codes and Statues

The following are API codes and their definitions.

Code	Status	Description
200	OK	The request was successfully completed.
201	Created	A new resource was successfully created.
400	Bad Request	The request was invalid.
401	Unauthorized	The request did not include an authentication token, or the authentication token was expired.
403	Forbidden	The client did not have permission to access the requested resource.
404	Not Found	The requested resource was not found.
405	Method Not Allowed	The resource did not support the HTTP method in the request. For example, the DELETE method cannot be used with the Agent API.

409	Conflict	The request could not be completed due to a conflict. For example, POST ContentStore Folder API cannot complete if the given file or folder name exists in the parent location.
500	Internal Server Error	The request was not completed due to an internal error on the server-side.
503	Service Unavailable	The server was unavailable.

4. Using API in Python

Lets have a fun with StackOverFlow API. Open python IDE of your choice. And copy the following code.

Open Pycharm, New Script and run the following code

```
import json
import requests

response=
requests.get('https://api.stackexchange.com/2.3/questions?order=desc&sort=activity&site=s
tackoverflow')

print(response.json()['items'])
```

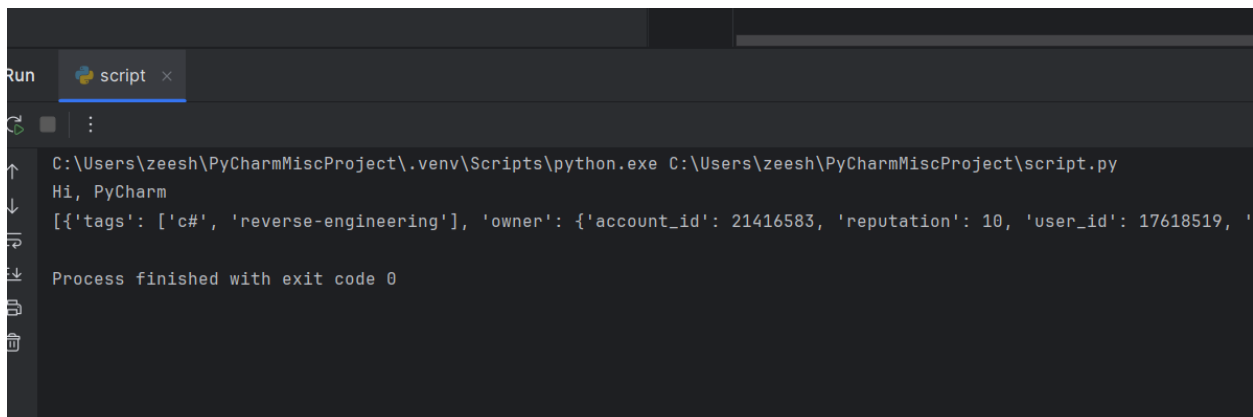
Note: you might need to install requests. To do that, open the python terminal and copy the following:

```
$ python -m pip install requests
```

Your code should look like this.

```
# -*- coding: utf-8 -*-  
"""  
Created on Mon Aug 30 22:59:19 2021  
  
@author: User1  
"""  
  
import json  
import requests  
  
response = requests.get('https://api.stackexchange.com/2.3/questions?order=desc&sort=activity&site=stackoverflow')  
print(response.json()['items'])
```

The output should be like this.



```
Run script x  
C:\Users\zeesh\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\zeesh\PyCharmMiscProject\script.py  
Hi, PyCharm  
[{'tags': ['c#', 'reverse-engineering'], 'owner': {'account_id': 21416583, 'reputation': 10, 'user_id': 17618519, '  
Process finished with exit code 0
```

And if you copy it in notepad, it would look like this:

```
[{'tags': ['url', 'amazon-route53', 'nginx-reverse-proxy', 'shiny-server'], 'owner': {'account_id': 19911270, 'reputation': 3, 'user_id': 14589238, 'user_type': 'registered', 'profile_image': 'https://www.gravatar.com/avatar/d615453debf3c06b201b338c7a8f7345?s=128&d=identicon&r=PG&f=1', 'display_name': 'maillo', 'link': 'https://stackoverflow.com/users/14589238/maillo'}, 'is_answered': True, 'view_count': 26, 'accepted_answer_id': 68970592, 'answer_count': 2, 'score': 0, 'last_activity_date': 1630360827, 'creation_date': 1630219343, 'last_edit_date': 1630331163, 'question_id': 68970517, 'content_license': 'CC BY-SA 4.0', 'link': 'https://stackoverflow.com/questions/68970517/how-to-define-one-domain-for-both-static-webpage-and-shiny-server-app'}, {'tags': ['sql', 'postgresql', 'triggers'], 'owner': {'account_id': 19507317, 'reputation': 9, 'user_id': 14574710, 'user_type': 'registered', 'profile_image': 'https://www.gravatar.com/avatar/5a7ec7fa8601227d1396c0465e7254d7?s=128&d=identicon&r=PG&f=1', 'display_name': 'Mark', 'link': 'https://stackoverflow.com/users/14574710/mark'}, 'is_answered': False, 'view_count': 31, 'answer_count': 2, 'score': -2, 'last_activity_date': 1630360814, 'creation_date': 1630203944, 'last_edit_date': 1630221937, 'question_id': 68969443, 'content_license': 'CC BY-SA 4.0', 'link': 'https://stackoverflow.com/questions/68969443/erro-gatilhos-postgresql', 'title': 'Erro gatilhos PostgreSQL'}, {'tags': ['python', 'getattr'], 'owner': {'account_id': 19234758, 'reputation': 11, 'user_id': 14055933, 'user_type': 'registered', 'profile_image': 'https://www.gravatar.com/avatar/881dd6190618798fb13b7a89eff58790?s=128&d=identicon&r=PG&f=1', 'display_name': 'snarayanan', 'link': 'https://stackoverflow.com/users/14055933/snarayanan'}, 'is_answered': False, 'view_count': 2, 'answer_count': 0, 'score': 0, 'last_activity_date': 1630360814, 'creation_date': 1630360814, 'question_id': 68990768, 'content_license': 'CC BY-SA 4.0', 'link': 'https://stackoverflow.com/questions/68990768/why-does-getattr-catch-all-attributeerrors-and-not-just-the-attribute-its-looki', 'title': 'Why does getattr catch all AttributeErrors and not just the attribute it&#39;s looking for?'}, {'tags': ['html', 'css'], 'owner': {'account_id': 19913763, 'reputation': 1, 'user_id': 16790997, 'user_type': 'registered', 'profile_image': 'https://lh3.googleusercontent.com/a-/AOh14Gh92sV8P9xMip0-FXtmeev05Ey13ASHAsMni-zIOg-k-s128', 'display_name': 'TGP1cccul8r', 'link': 'https://stackoverflow.com/users/16790997/tgp1cccul8r'}, 'is_answered': False, 'view_count': 19, 'answer_count': 1, 'score': 0, 'last_activity_date': 1630360812, 'creation_date': 1630359082, 'last_edit_date': 1630360711, 'question_id': 68990551, 'content_license': 'CC BY-SA 4.0', 'link': 'https://stackoverflow.com/questions/68990551/how-to-get-an-image-to-go-to-a-corner-and-automatically-adjust-its-size', 'title': 'How to get an image to go to a corner and automatically adjust its size'}, {'tags': ['python', 'uml', 'pylint'], 'owner': {'account_id': 1317462, 'reputation': 9733, 'user_id': 1264018, 'user_type': 'registered', 'accept_rate': 81, 'profile_image': 'https://www.gravatar.com/avatar/c0338476eea2e564d77246b7234773e1?s=128&d=identicon&r=PG&f=1', 'display_name': 'feelfree', 'link': 'https://stackoverflow.com/users/1264018/feelfree'}, 'is_answered': True, 'view_count': 23, 'accepted_answer_id': 68987719, 'answer_count': 1, 'score': 2, 'last_activity_date': 1630360809, 'creation_date': 1630260359, 'question_id': 68975663, 'content_license': 'CC BY-SA 4.0', 'link': 'https://stackoverflow.com/questions/68975663/can-python-pyreverse-generate-uml-diagram-for-functions', 'title': 'Can Python pyreverse generate UML diagram for functions?'}, {'tags': ['java', 'spring-boot', 'elasticsearch', 'spring-data-elasticsearch'], 'owner': {'account_id': 13010042, 'reputation': 529, 'user_id': 9403408, 'user_type': 'registered', 'accept_rate': 100, 'profile_image': 'https://www.gravatar.com/avatar/0163d1cb80cfa8743fc2fba743beead4?s=128&d=identicon&r=PG&f=1', 'display_name': 'H Athukorala', 'link': 'https://stackoverflow.com/users/9403408/h-athukorala'}, 'is_answered': False, 'view_count': 6, 'answer_count': 0, 'score': 0, 'last_activity_date': 1630360804, 'creation_date': 1630355364, 'last_edit_date': 1630360804, 'question_id': 68989964, 'content_license': 'CC BY-SA 4.0', 'link': 'https://stackoverflow.com/questions/68989964/how-to-search-by-nested-field-id-using-nativesearchquerybuilder-in-elasticsearch', 'title': 'How to search by nested field id using NativeSearchQueryBuilder in ElasticsearchRepository?'}, {'tags': ['videochat', 'video-conferencing', 'laragon', 'videocall'], 'owner': {'account_id': 20228932, 'reputation': 7, 'user_id': 14836032, 'user_type': 'registered', 'profile_image': 'https://lh3.googleusercontent.com/a-/
```

Now try the following code:

```
import json
import requests

response=
requests.get('https://api.stackexchange.com/2.3/questions?order=desc&sort=activity&site=stackoverflow')

for data in response.json()['items']:
    if data['answer_count']==0:
        print(data['title'])
        print(data['link'])
    else:
        print('skipped')
    print()
```

This code looks for un answer questions in the JSON array. Furthermore, return the question title and the link for the question. The output should be liked the following snippet.


```
codebyte Challenge: Questions Marks
https://stackoverflow.com/questions/68990907/codebyte-challenge-questions-marks

skipped

skipped

skipped

skipped

skipped

Starting tmux on Google cloud
https://stackoverflow.com/questions/68990902/starting-tmux-on-google-cloud

skipped

skipped

skipped

skipped

Visual Studio Python Kivy Unknown Directive Error
https://stackoverflow.com/questions/68990898/visual-studio-python-kivy-unknown-directive-error

why is not being calculated my Conditional Value at Risk code properly and sometimes calculation takes too long 4000ms in TradingView
https://stackoverflow.com/questions/68747860/why-is-not-being-calculated-my-conditional-value-at-risk-code-properly-and-somet

skipped

skipped

Value is not updating with custom select component
https://stackoverflow.com/questions/68990776/value-is-not-updating-with-custom-select-component

Unable to use nested window functions in Looker
https://stackoverflow.com/questions/68990895/unable-to-use-nested-window-functions-in-looker
```

At this point, you should be familiar with the API and how to use it.

Part 2: Exploring data with Postman

Step 1: Familiarize yourself with Postman

We have already installed and use Postman for exploring StackOverflow API in an earlier section. However, this section will explore an API with Postman and familiarize ourselves with Postman key concepts and how to use it.

Requests

A request is what it sounds like; the request is a specific API request. It can be GET or POST to a specific endpoint.

Collections

A collection is a group of requests. It is convenient for organizing requests into different groups. This could be as simple as two different APIs (i.e. Twitter vs Slack) or two different groups of APIs for a single API (i.e. Twitter Tweets API vs Twitter Accounts API).

Authorization

Authorization is how requests are authenticated with an API, whether by a person making a request or by a computer making that request on your behalf. This commonly comes in the form of an API

key which can be a static value assigned to your account or dynamically generated with tools like OAuth.

Environments

Environments will allow you to configure your endpoints to use specific variables, making it easier to use the same endpoints between different environments. For instance, you might have the same **/profile** endpoint on your production and development environments, but they have different domains. Environments let you manage a single request with a variable domain.

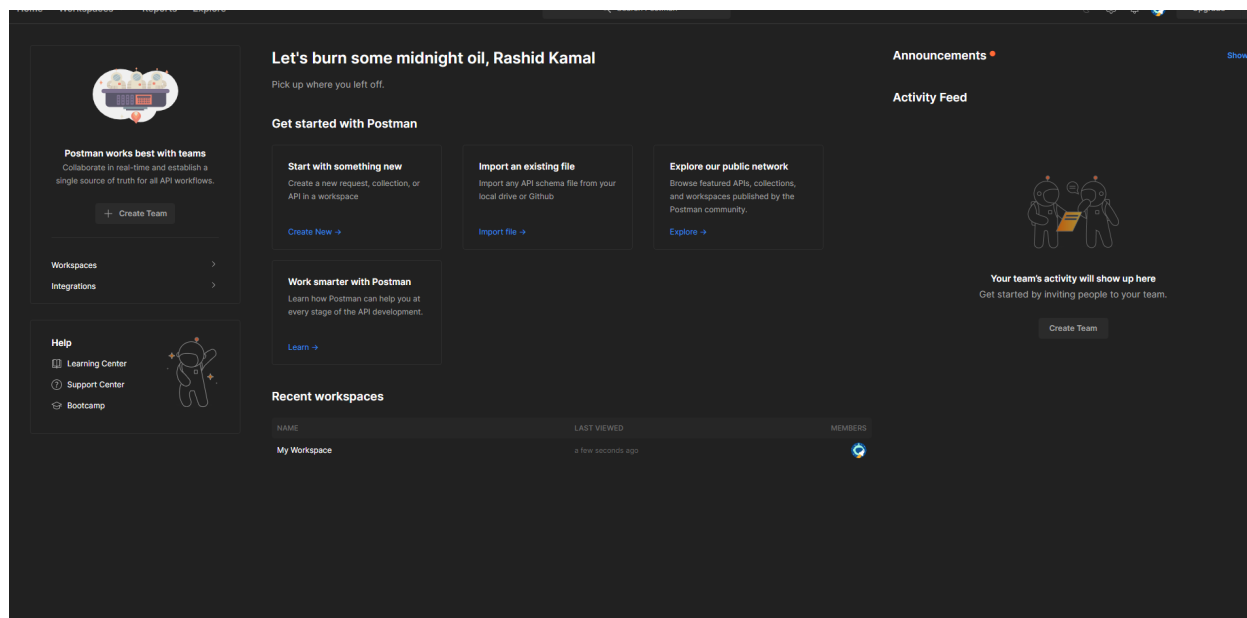
Workspaces

Workspaces allow you to manage and organize different sets of collections. Imagine if you want to use Postman for both work and a personal project, you might have a Work workspace and a Personal workspace.

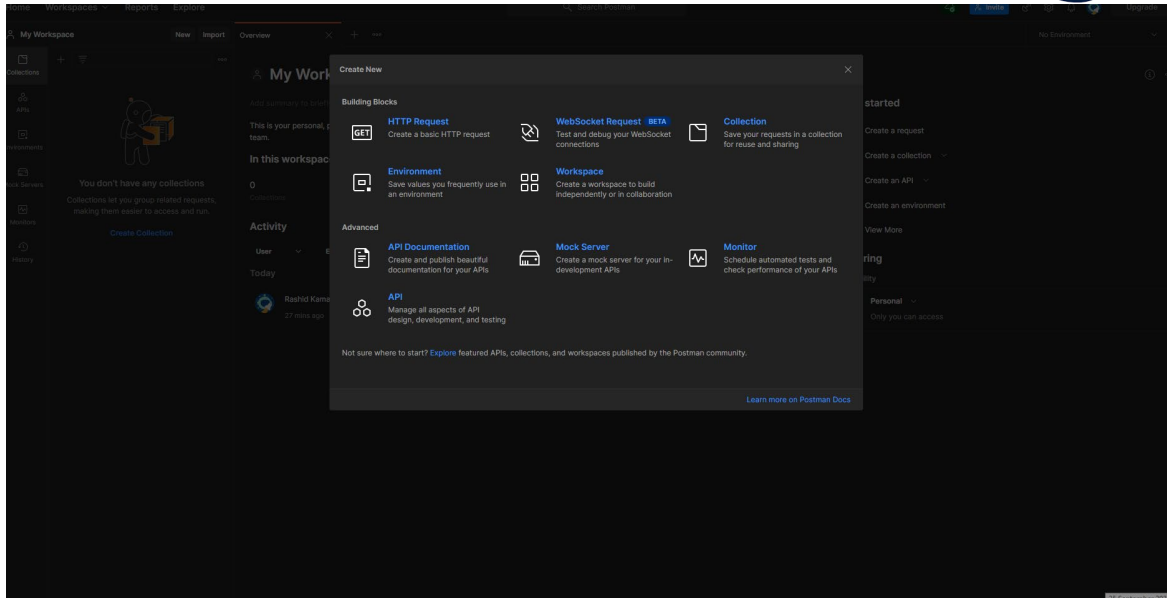
Step 2: Creating new Postman request to GET info about Squirtle

Please create an account in Postman. Some of the features are disabled, and you will need to create an account in Postman. You can sign up with your Google account.

Click on the home button and click on the first tab start with something new



Click on Create new tab



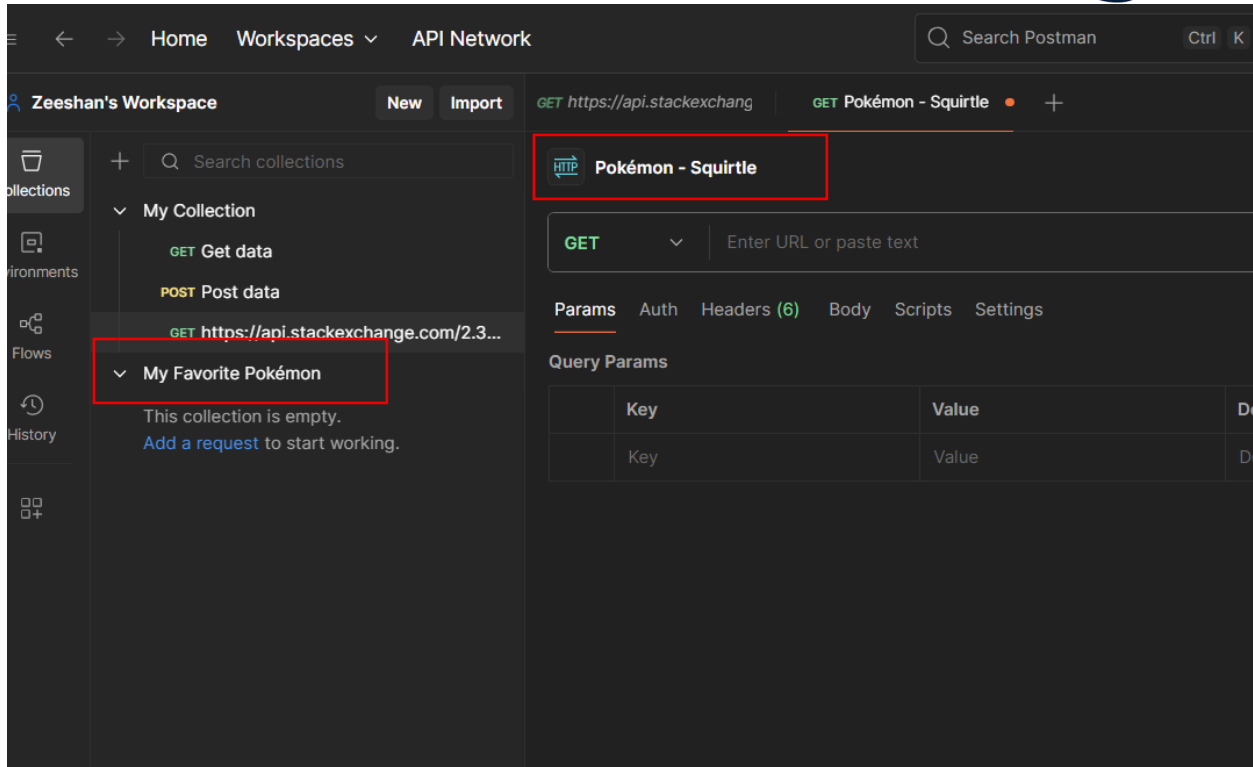
Click on the GET HTTP request tab.

Before we get into the request itself, it requests a few things.

The first thing requires a name. We're going to start by requesting information about the Pokémon Squirtle, so let's name this "Pokémon - Squirtle".

It also requires a collection, so click Create Collection and let's name the collection "My Favorite Pokémon".

Click on the + button at the top of the page.

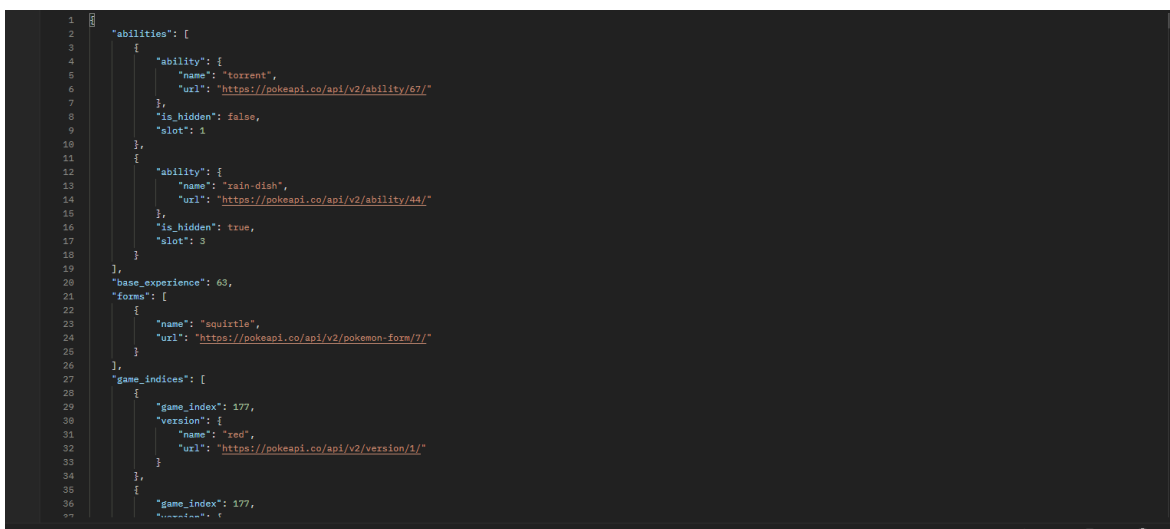


There are two things we'll first need to fill out for our first request:

Request type: GET, POST, PUT, etc. - we will use GET

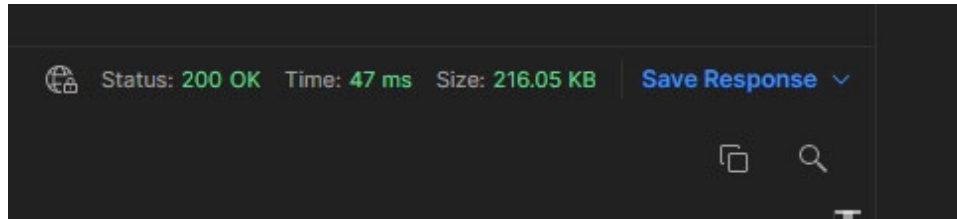
Request URL: The endpoint for your API request - for our request, we will use <https://pokeapi.co/api/v2/pokemon/squirtle/>

Copy the URL and click on the send button.



You should see the return JSON data.

On the left side of the data panel, you should see request status too.



Body: At the bottom, we should now see the response body of the API request. We should have a JSON object with data like abilities, base_experience, and forms for our Squirtle API.

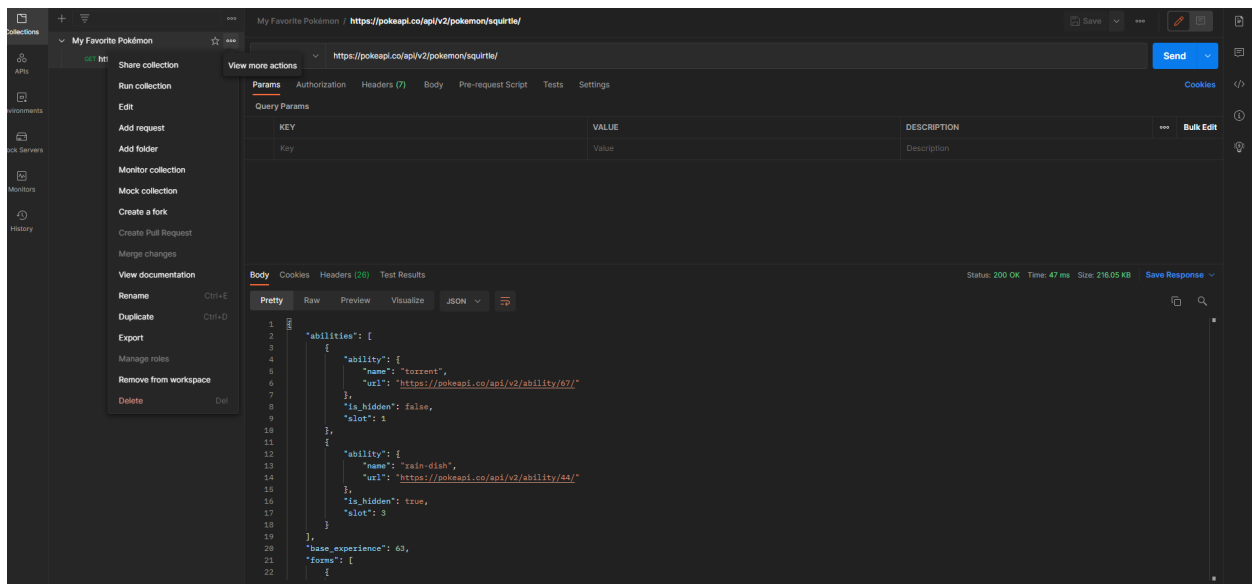
Status: on the right, we should see the HTTP status code. "200 OK" is a good sign, and it means it was successful! (The complete HTTP status Code Table is mentioned in the first section)

Time: how long the request took to finish

Size: the size in KB (in our example) of the response data

Save the file in the collection we just created.

Click on **view more actions** and add a new folder and named Pokemon.



Click on **view more actions** in the new folder you just created.

Click on **add request** name the request Pokemon too.

Request Type: GET

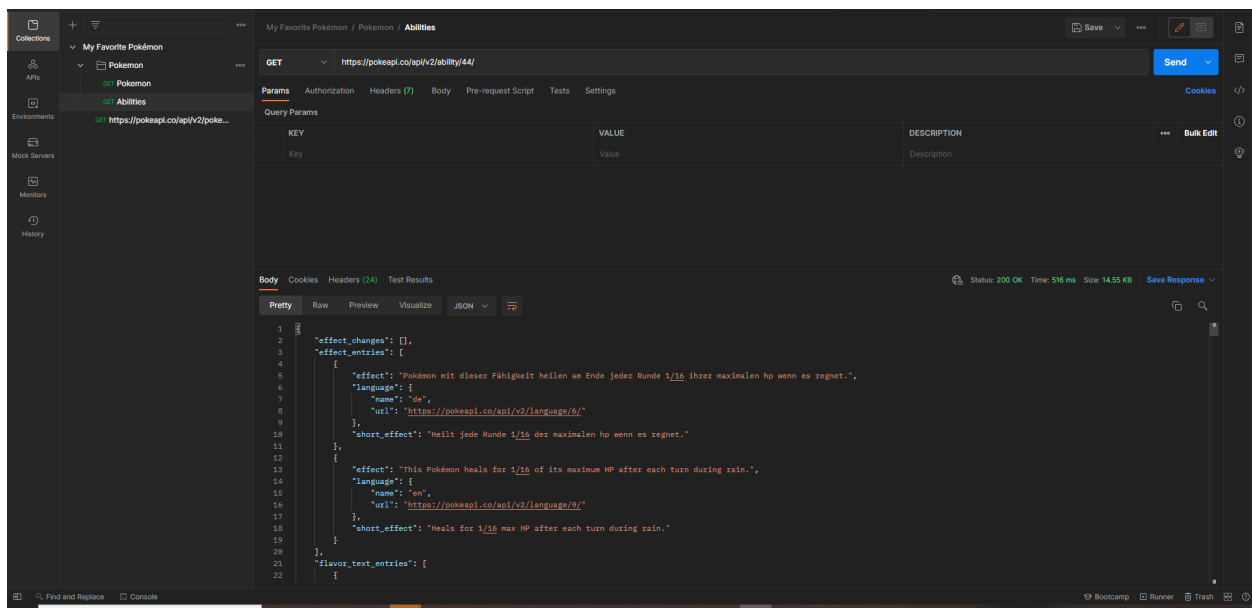
Request URL: <https://pokeapi.co/api/v2/pokemon/squirtle/>

Now let's add another request. Follow the same process as before to create a new request under the PokéAPI Pokémon folder and let's name this request "Abilities".

If you scroll through the response from the first Squirtle endpoint, you see a lot of other API urls. At the top, we have abilities and two different ones — "torrent" and "rain-dish".

Choose your favourite Squirtle ability and copy the url value into the new Abilities request we just created; I'm going to use **rain-dish**.

We can leave the Request Type as GET, hit the blue Send button, and see a successful response again!



Here we get a lot of information about our Squirtle ability Rain Dish and some of the details come in different languages, which is excellent!

So now we have a new PokéAPI collection with a Pokémon folder representing Pokémon API endpoints, including Pokemon and abilities.

We're going to stop this section with those two requests, but feel free to continue and add as many of the PokéAPI requests as you'd like!

Step 3: Making POST requests with Postman to translate sentences to sound like Yoda

So far we've only made GET requests, but what if we wanted to make a POST request to send some data?

For making a POST request, we're going to use the Yoda Translator API from funtranslations.com. While this API only takes a single parameter, it's still an excellent public endpoint we can use to understand the concept.

First, let's create a new collection with a new request:

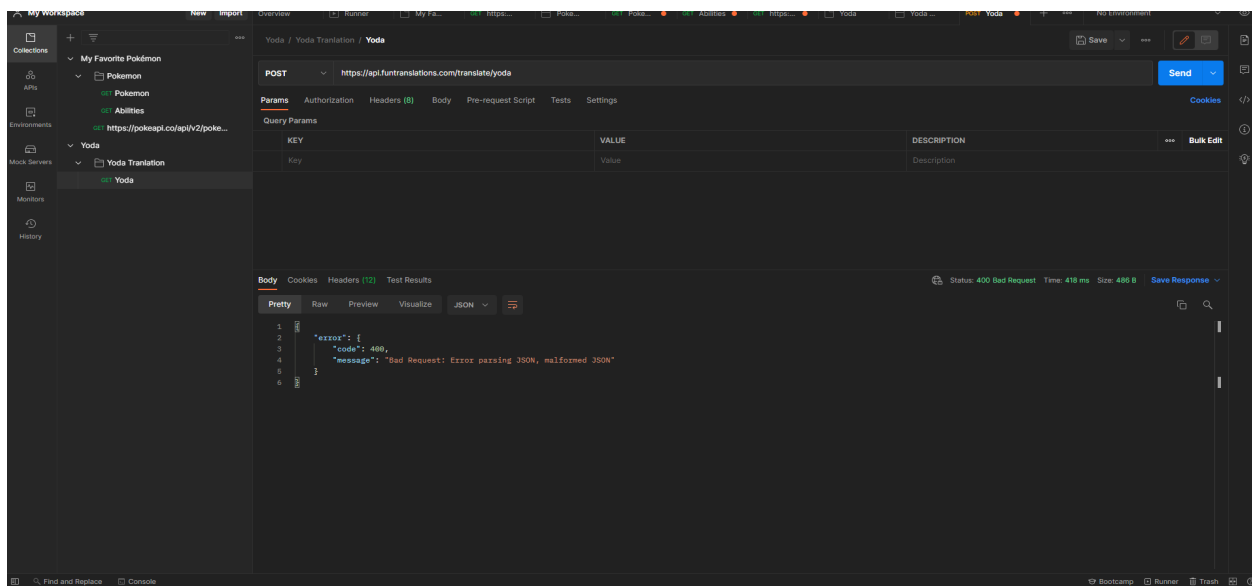
Collection: Fun Translations

Request: Yoda

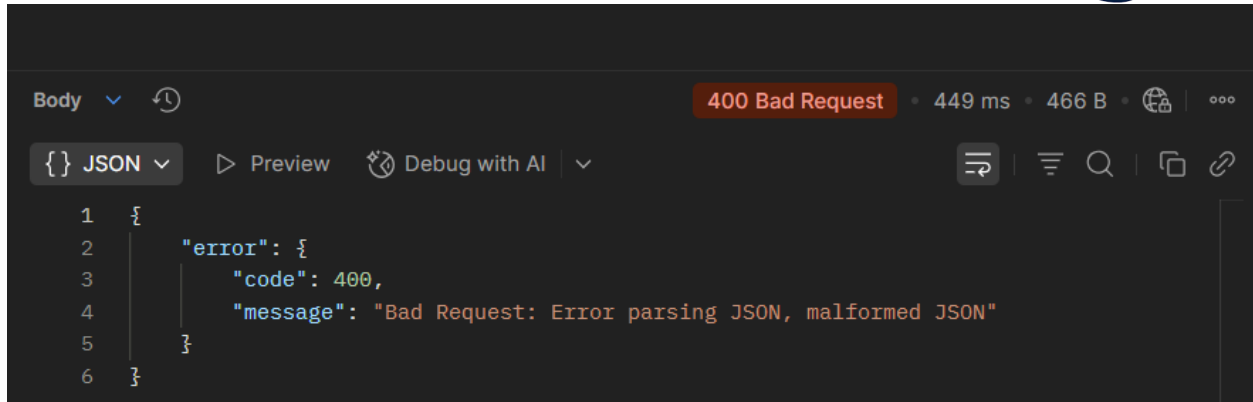
This time, instead of a GET request, our request configuration will be:

Request Type: POST

Request URL: <https://api.funtranslations.com/translate/yoda>

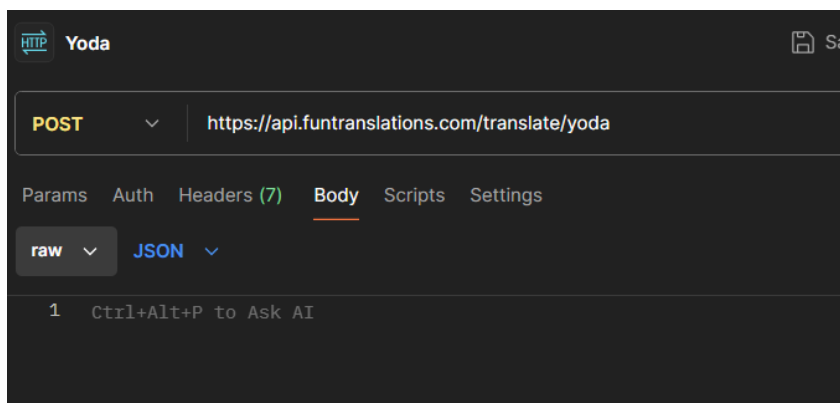


Now this time, if we hit the blue Send button, we'll notice we don't get a successful 200 response, we get a **400!**



We never actually set up any data to be posted to the API, and it requires that data, so let's add it.

Right below the Request URL, click Body. Then instead of none, select raw as the body type. Finally, on the right of the types, change Text to JSON.

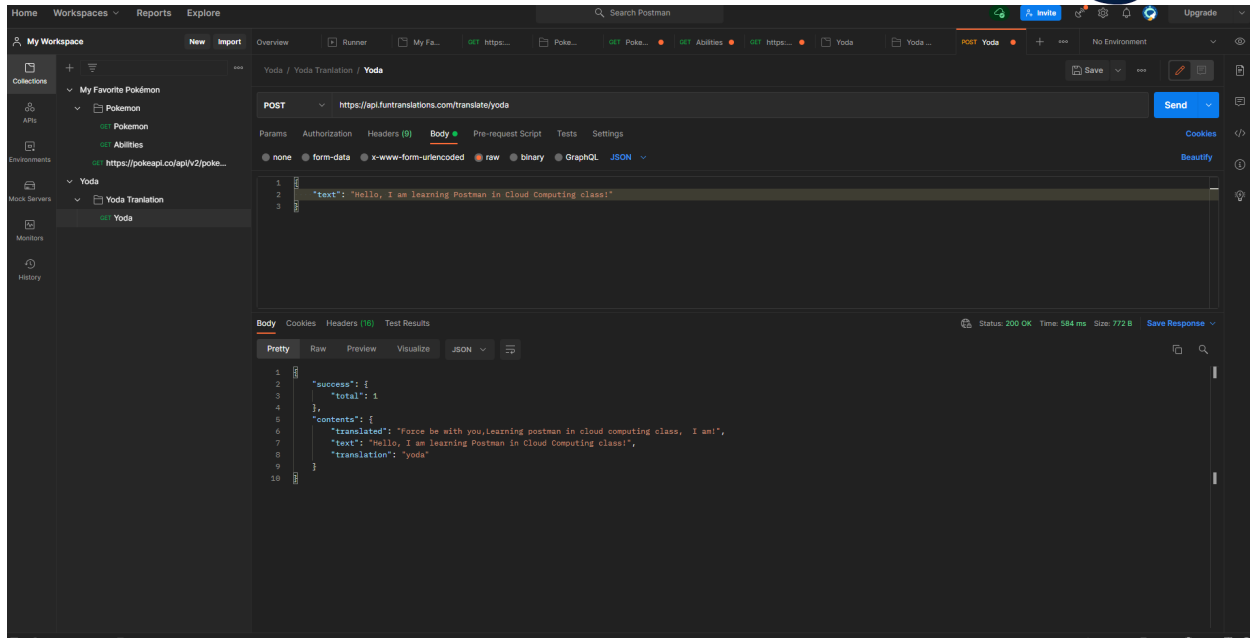


Now type any text you would like to translate.

```

{
  "text": "Hello, I am learning Postman in Cloud Computing class!"
}

```

We can apply this concept to pretty much any API. Postman doesn't only permit you to post JSON, it allows you to use the other formats that we see listed in the Body Type section, meaning you have a lot of options depending on what the API you're using requires.

Part 3: Host a RESTful API in Azure App Service

[Azure App Service](#) provides a highly scalable, self-patching web hosting service. This part will show how to deploy an ASP.NET Core API app to App Service. You configure the app using command-line tools and deploy the app using Git.

Create local ASP.NET Core app

In this step, you set up the local ASP.NET Core project. App Service supports the same workflow for APIs written in other languages.

Clone the sample application

- Open Windows PowerShell and cd to a working directory.
- Clone the sample repository and change it to the repository root by copying the following code and past it into the PowerShell.

```
git clone https://github.com/Azure-Samples/dotnet-core-api
```

```
cd dotnet-core-api
```

 Administrator: Windows PowerShell

```
PS C:\> git clone https://github.com/Azure-Samples/dotnet-core-api
>> cd dotnet-core-api
Cloning into 'dotnet-core-api'...
remote: Enumerating objects: 82, done.
remote: Total 82 (delta 0), reused 0 (delta 0), pack-reused 82
Unpacking objects: 100% (82/82), done.
PS C:\dotnet-core-api>
```

This repository contains an app that's created based on the following tutorial: [ASP.NET Core Web API help pages using Swagger](#). It uses a Swagger generator to serve the [Swagger UI](#) and the Swagger JSON endpoint.

Make sure the default branch is main.

```
git branch -m main
```

 Administrator: Windows PowerShell

```
PS C:\> git clone https://github.com/Azure-Samples/dotnet-core-api
>> cd dotnet-core-api
Cloning into 'dotnet-core-api'...
remote: Enumerating objects: 82, done.
remote: Total 82 (delta 0), reused 0 (delta 0), pack-reused 82
Unpacking objects: 100% (82/82), done.
PS C:\dotnet-core-api> git branch -m main
PS C:\dotnet-core-api>
```

Run the application

- c) Run the following commands to install the required packages, run database migrations, and start the application.

If you encounter the following error, please install Dotnet by using following link:

- <https://dotnet.microsoft.com/en-us/download/dotnet/9.0>

Note: You should have restarted PowerShell or terminal after installing the .NET Core 3.1 SDK.

- d) Run the .Net by copying the following code into PowerShell.

dotnet restore

dotnet

run

```
https://api.nuget.org/v3/index.json
PS C:\Users\zeesh\Documents\dotnet-core-api> dotnet restore
Restore complete (22.6s)

Build succeeded in 23.8s
```

```
Content root path: C:\dotnet-core-api
info: Microsoft.Hosting.Lifetime[0]
      Application is shutting down...
PS C:\dotnet-core-api> dotnet run
info: Microsoft.Hosting.Lifetime[0]
      Now listening on: http://localhost:5000
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\dotnet-core-api
```

If you see an error, do following:

Run this command in your project folder:

```
1 dotnet nuget list source
```

Ensure you see:

```
nuget.org https://api.nuget.org/v3/index.json
```

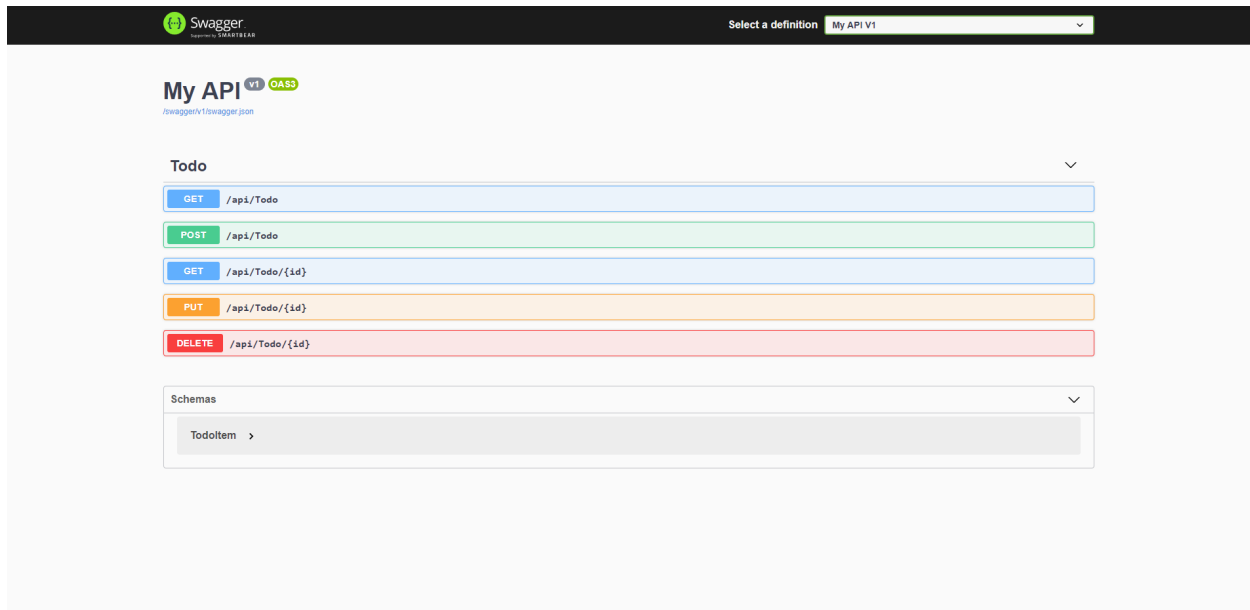
If not, add it:

```
1 dotnet nuget add source https://api.nuget.org/v3/index.json -n nuget.org
```

- e) Now go to the browser and put the following link. You must not terminate the currently running instance in the PowerShell.

<http://localhost:5000/swagger/index.html>

You should see the following snippet in your browser.



- f) Navigate to <http://localhost:5000/api/todo> and see a list of ToDo JSON items.
- g) Navigate to <http://localhost:5000> and play with the browser app. Later, you will point the browser app to a remote API in App Service to test CORS functionality. Code for the browser app is found in the repository's *wwwroot* directory.
- h) To stop ASP.NET Core at any time, press Ctrl+C in the terminal.

Use Azure Cloud Shell

Azure hosts Azure Cloud Shell, an interactive shell environment that you can use through your browser. You can use either Bash or PowerShell with Cloud Shell to work with Azure services.

Go to the following link.

<https://portal.azure.com/#cloudshell/>



Microsoft Azure

Search resources, services, and docs (G+J)

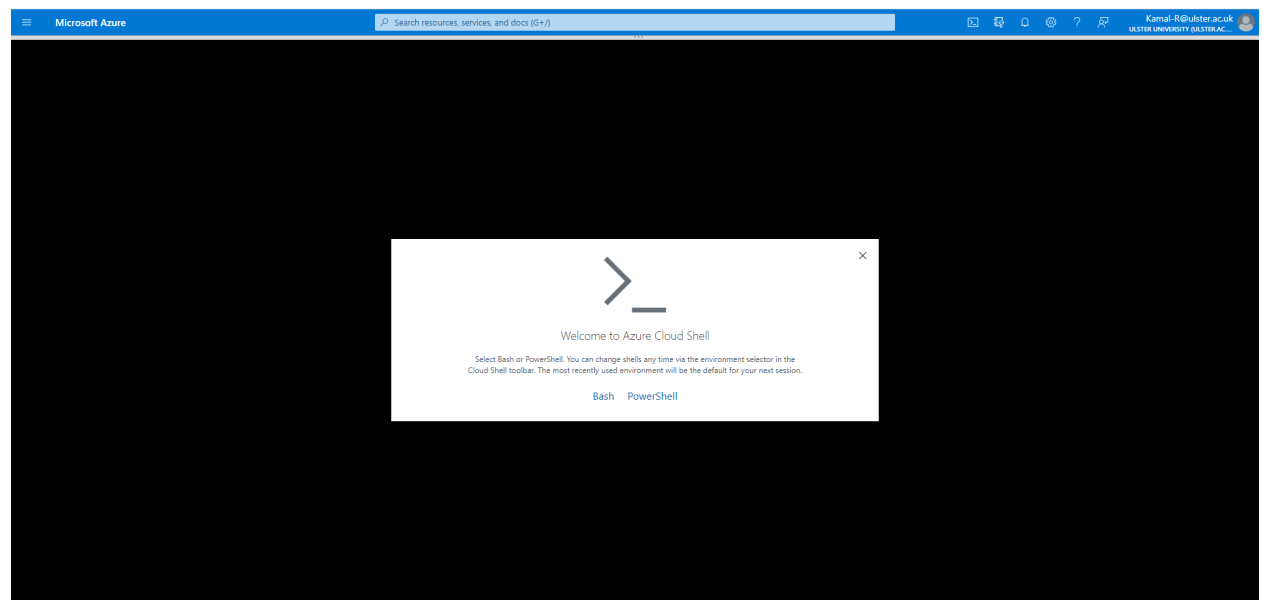
Azure services

- Create a resource
- App Services
- Resource groups
- Education
- Connections
- Virtual networks
- Virtual machines
- SQL databases
- Free services
- More services

Resources

Recent Favorite

Name	Type	Last Viewed
UlsterAzureExample	App Service	22 hours ago
mySampleDatabase	SQL database	23 hours ago



Click on either Bash or PowerShell.

Your screen will pop up with the following snippet.

```
PowerShell ▾ | ○ ? ⓘ ⏏ ⏏ ⏏ ⏏ ⏏
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

MOTD: Customize your experience: save your profile to $HOME/.config/PowerShell

VERBOSE: Authenticating to Azure ...
VERBOSE: Building your Azure drive ...
PS /home/rashid
```

1. `git clone https://github.com/Azure-Samples/dotnet-core-api`
2. `cd dotnet-core-api`

```
Powershell | ? @ # $ % & ' " { } |
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

MOVD: Customize your experience: save your profile to $HOME/.config/Powershell

VERBOSE: Authenticating to Azure ...
VERBOSE: Building your Azure drive ...
PS /home/rashid> git clone https://github.com/Azure-Samples/dotnet-core-api
Cloning into 'dotnet-core-api'...
remote: Enumerating objects: 82, done.
remote: Total 82 (delta 0), reused 0 (delta 0), pack-reused 82
Unpacking objects: 100% (82/82), done.
PS /home/rashid> cd dotnet-core-api
PS /home/rashid/dotnet-core-api>
```

Deploy the app to Azure

In this step, you deploy your SQL Database-connected .NET Core application to App Service.

Configure local git deployment

FTP and local Git can deploy to an Azure web app by using a *deployment user*. Once you configure your deployment user, you can use it for all your Azure deployments. Your account-level deployment username and password are different from your Azure subscription credentials.

To configure the deployment user, run the [az webapp deployment user set](#) command in Azure Cloud Shell. Replace <username> and <password> with a deployment user username and password.

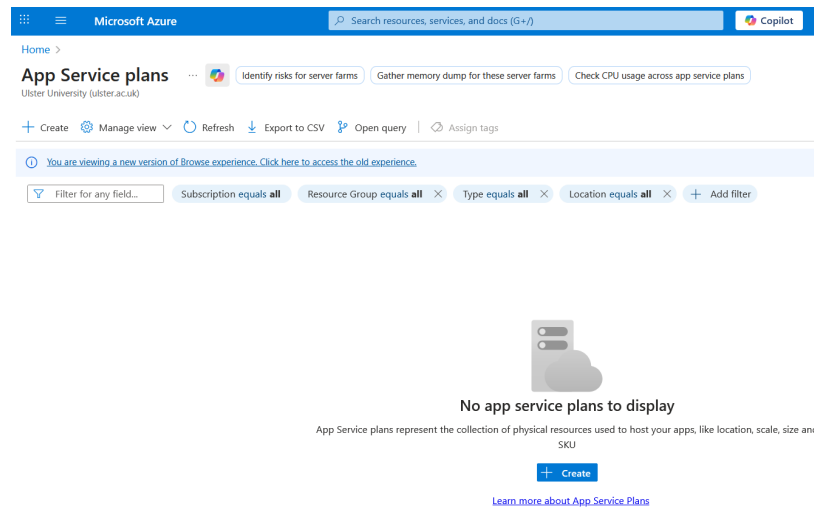
- The username must be unique within Azure, and for local Git pushes, it must not contain the '@' symbol.
 - The password must be at least eight characters long, with two of the following three elements: letters, numbers, and symbols.
- `az webapp deployment user set --user-name <username> --password <password>`

```
PS /home/rashid/dotnet-core-api> az webapp deployment user set --user-name cloudy15 --password us123123
{
  "id": null,
  "kind": null,
  "name": "web",
  "publishingPassword": null,
  "publishingPasswordHash": null,
  "publishingPasswordHashSalt": null,
  "publishingUserName": "cloudy15",
  "scmUri": null,
  "systemData": null,
  "type": "Microsoft.Web/publishingUsers/web"
}
PS /home/rashid/dotnet-core-api>
PS /home/rashid/dotnet-core-api>
```

The JSON output shows the password is null if you get a 'Conflict'. Details: 409 error, change the username. If you get a 'Bad Request'. Details: 400 error, use a stronger password.

Record your username and password to use to deploy your web apps.

Create an App Service plan (and a resource group)



Home > App Service plans >

Create App Service Plan ...

Subscription * ⓘ Azure for Students

Resource Group * ⓘ (New) myResourceGroup
[Create new](#)

App Service Plan details

Name * myAppServicePlan ✓

Operating System * ☒ Linux ☐ Windows

Region * Canada Central

Pricing Tier

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app.
[Learn more](#)

Pricing plan Free F1 (Shared infrastructure)
[Explore pricing plans](#)

Create new
resource
group

This should
be the
region from
your policy
(see lab 4
for details)

Zone redundancy

An App Service plan can be deployed as a zone redundant service in the regions that support it. Your initial instance count will be set based on your zone redundancy configuration. To ensure you'll be able to enable zone redundancy at any point in the lifecycle of your app, enable zone redundancy now. You can decrease the instance count after the App Service plan is created. [Learn more](#)

i Zone redundancy is not available on the selected pricing tier.

[Review + create](#)

[< Previous](#)

[Next : Tags >](#)

Microsoft Azure

Search resources, services, and docs (G+/)

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Microsoft.Web-ASP-Portal-a4c12e12-9ce2 | Overview ...

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.Web-ASP-Portal-a4c12e12-9ce2

Subscription : [Azure for Students](#)

Resource group : [myResourceGroup](#)

Start time : 10/19/2025, 10:06:00 AM

Correlation ID : bd3d1783-b4f0-4f57-821c-4ec752a73648

> Deployment details

✓ Next steps

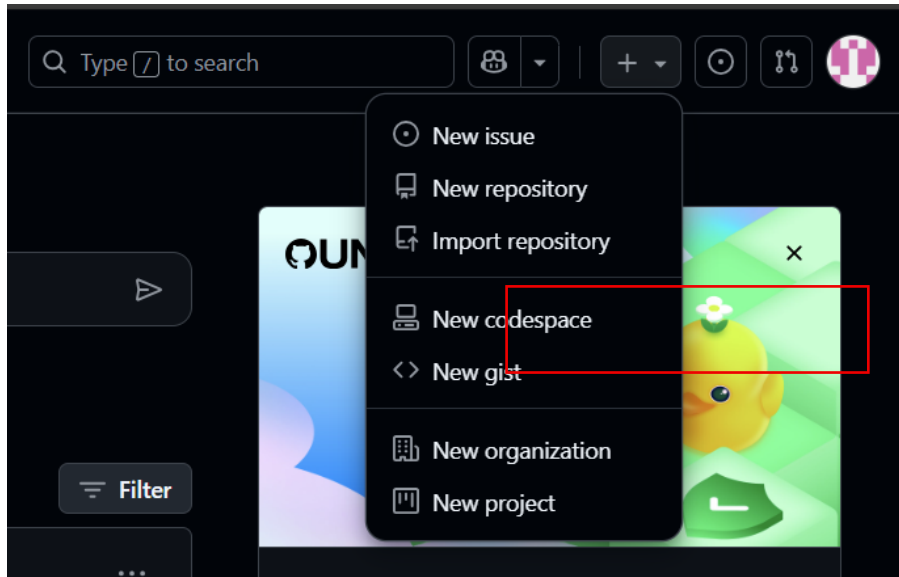
Go to resource

Create a web app on Azure

In this step we will:

1. Link the Git public repo of dotnet-core-api to your personal Github account.
2. Create a webapp at Azure
3. Link your Azure app with your Git repository.
4. Test the application

First login to your github account and click 'import repository'



Enter this link and add the repository name <https://github.com/Azure-Samples/dotnet-core-api>.

Required fields are marked with an asterisk ().*

Support for importing Mercurial, Subversion and Team Foundation Version Control (TFVC) repositories ended on April 12, 2024. For more details, see the [changelog](#).

Your source repository details

The URL for your source repository *

Learn more about [importing git repositories](#).

Please enter your credentials if required for cloning your remote repository.

Your username for your source repository

Your access token or password for your source repository

Your new repository details

Owner *

 zeeshanads

Repository name *

✔ dotnetapp2 is available.

☐ Public

Create webapp by searching App Services, create new Webapp

[Home](#) > [App Services](#) >

Create Web App ...

Subscription * ⓘ

Azure for Students

Resource Group * ⓘ

(New) Azurewebappcom682_group

[Create new](#)

Instance Details

Name

Azurewebappcom682

.azurewebsites.net



Secure unique default hostname on. [More about this update](#)

Publish *



Code



Container

Runtime stack *

.NET 8 (LTS)

Operating System *



Linux



Windows

Region *

Norway East



Not finding your App Service Plan? Try a different region or select your App Service Environment.

Pricing plans

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app.

[Learn more](#)

Windows Plan (Norway East) * ⓘ

(New) MyServiceAppPlan

[Create new](#)

Pricing plan

Free F1 (Shared infrastructure)

[Review + create](#)

[< Previous](#)

[Next : Database >](#)

On deployment page, enter your GitHub account details, enter your GitHub organisation, and use drop down to select the repository that you have just created. In Brach option use master.

[Home](#) > [App Services](#) >

Create Web App ...

Basics Database **Deployment** Networking Monitor + secure Tags Review + create

Continuous deployment settings

Set up continuous deployment to easily deploy code from your GitHub repository via GitHub Actions. [Learn more](#)

Continuous deployment ☐ Disable ☒ Enable

GitHub settings

Set up GitHub Actions to push content to your app whenever there are code changes made to your repository. Note: Your GitHub account must have write access to the selected repository in order to add a workflow file which manages deployments to your app.

GitHub account	zeeshanads
	Change account ⓘ
Organization *	zeeshanads
Repository *	dotnet
Branch *	master

Workflow configuration

Click the button below to preview what the GitHub Actions workflow file will look like before setting up continuous deployment.

[Preview file](#)

Authentication settings

[Review + create](#)

[< Previous](#)

[Next : Networking >](#)

Click on create

Microsoft Azure

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Home >

Microsoft.Web-WebApp-Portal-ba70fa67-b8d0 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

Deployment is in progress

Deployment name : Microsoft.Web-WebApp-Portal-ba70fa67-b8d0 Start time : 10/19/2025, 12:10:26 PM
 Subscription : [Azure for Students](#) Correlation ID : d45c372a-0d14-4e32-9be6-c72ed89eae8
 Resource group : [Azurewebappcom682_group](#)

Deployment details

Resource	Type	Status	Operation details
There are no resources to display.			

Browse Stop Swap Restart Delete Refresh Download publish profile Reset publish profile Share to mobile Send us your feedback

Enable Application Insights for monitoring and profiling for your app.

Essentials

Resource group (move) : [Azurewebappcom682_group](#)

Status : Running

Location (move) : Norway East

Subscription (move) : [Azure for Students](#)

Subscription ID : fa81128f-08de-4661-8799-706f7a129543

Tags (edit) : [Add tags](#)

Default domain : [azurewebappcom682-cac7hgakfec5gzht.norwayeast-01.azure...](#) [JSON View](#)

App Service Plan : [MyServiceAppPlan](#)

Operating System : Windows

Health Check : [Not Configured](#)

GitHub Project : <https://github.com/zeeshanads/dotnet>

Properties Monitoring Logs Capabilities Notifications (1) Recommendations

Here is the webpage that you will see:

azurewebappcom682-cac7hgakfec5gzht.norwayeast-01.azurewebsites.net/#/Home

Azure App Service REST API Sample

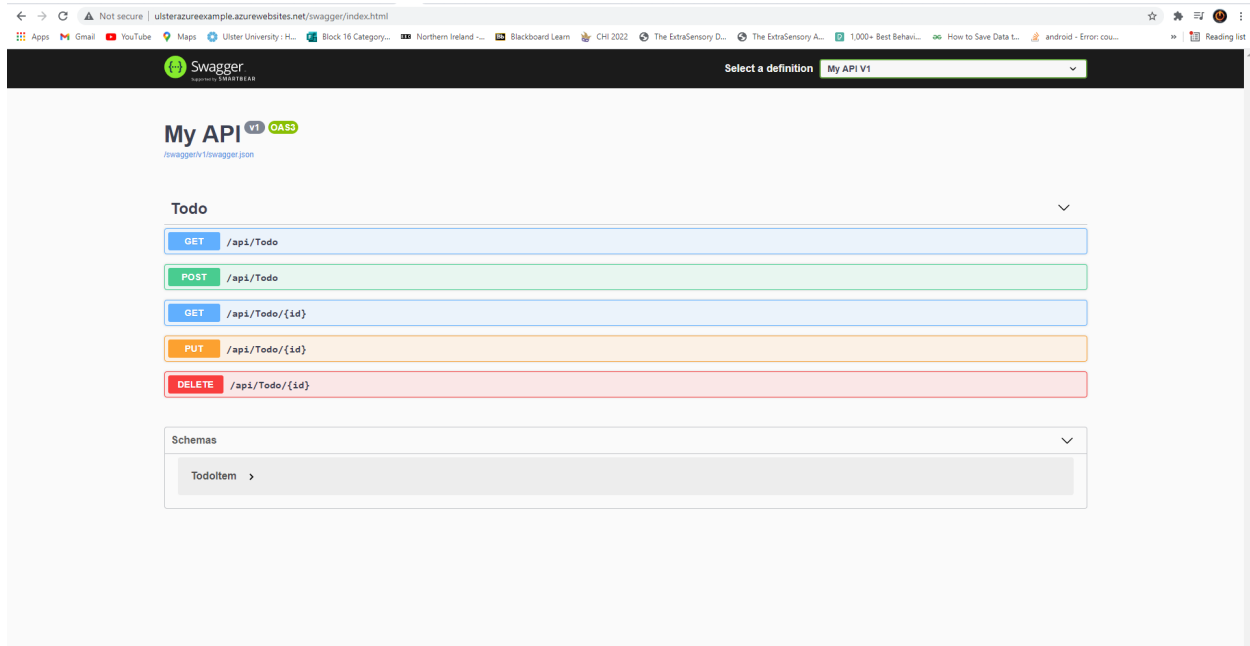
Todo List

☐ Item1	Edit Delete
☐	Edit Delete

Browse to the Azure app

<http://ulsterazureexample.azurewebsites.net/swagger/index.html>

You can access swagger index.html, by just adding /swagger/index.html at the end of the link you have copied from Default domain:



Congratulations: You have successfully deployed .NET API on your Azure Cloud Host.

In the end, you must delete the resource group and all other resources created in this lab.